

RE-MASTERS (RM.Sc.) IN COMPUTER SCIENCE
HWP LABS, BENIN CITY, EDO STATE
PROSPECTUS 2018/2019 SESSION

PREFACE

Objective

RMSC is a programme developed by HWP Labs to equip M.Sc. in Computer Science students with industry-relevant skills in Advance Software Engineering. At the end of the programme, students should be able to create full-stack web applications using JavaScript as well as apply general concepts of programming to other information software development projects-- regardless of the programming language of choice.

Scope

The RMSC SlingshotJS framework (CSC819) covers selected topics in Computer Science such as; Advanced Data Structures, Artificial Intelligence (AI), Compiler Design, Computer Communications and Networking, Database Management System, Design and Analysis of Algorithms, Expert Systems, Human-Computer Interaction (HCI), Information Technology (IT), Operations Research, Pattern Recognition, Programming Languages, Software Engineering and Systems Analysis and Design.

Modules

- ❖ Concepts of programming with JavaScript (ECMAScript 6).
- ❖ Source code editing with Visual Studio Code.
- ❖ Advanced Data Structures with JavaScript Object Notation (JSON).
- ❖ Parallel Processing with Asynchronous JavaScript and XML (AJAX).
- ❖ Server-side Scripting with Node.js Run-time Environment.
- ❖ Document-oriented Database Management with MongoDB.
- ❖ Software Development Life Cycle (SDLC) using Waterfall Model and MVC.
- ❖ Software Prototyping with Hypertext Markup Language (HTML5).

Prerequisite

- ✓ Text Editor –Visual Studio Code [\[download\]](#)
- ✓ Web Browser – Google Chrome [\[download\]](#)
- ✓ JavaScript Run-time Environment - Node.js [\[download\]](#)
- ✓ Database Management System – MongoDB [\[download\]](#)

NOTE: All software packages are currently available for transfer via CD or USB.

COURSE OUTLINE

MODULE 1: Concepts of programming with JavaScript (ECMAScript 6)			
#	LECTURE	SCOPE	EXERCISE
1	Program Structure <u>Components of a Webpage</u> Tags Attributes Elements Shell ☐ Website Architecture	Programming Domains The Internet World Wide Web	alert.js “Hello, World!”
2	Standard I/O <u>Syntax and Semantics</u> Expressions and Statements Identifiers and Scopes Programming Conventions ☐ Basic Flowchart of a Computer Program ☐ Algorithm for Program Design using MVC	Human-Computer Interaction (HCI) Command-line Interface (CLI) Algorithms Formula Translation	prompt.js confirm.js console.js ... loan_calc.js pay_slip.js
3	<u>Operators</u> Arithmetic Operators Assignment Operators Comparison Operators Logical Operators Operator Precedence ($z = -b + a * c / f + j$) ? Unary, Binary and Ternary Operators	Euclidean Division Algebraic Equations Flip Bit Mutation Product-Moment Correlation	modulus.js exponent.js ternary.js ... precedence.js
4	<u>Data Types</u> Primitive Data Types Composite Data Types Abstract Data Types Type Checking (typeof x) Type Conversion (Type Casting)	Data Validation	typeof.js
5	<u>Data Structures</u> Stack Queue *Pointer &Alias Dictionary (Hash Map or Assoc. Array) ? Difference between Stacks and Queues ? Applications of Genetic Algorithms	Abstract Data Types Enumerated Type Number Parity Arithmetic Crossover (GA)	array.js enum.js dictionary.js record.js ... scores.js
6	Control Structures I	Grading Systems	if.js

	<u>Decision Statements (Branch)</u> If Statement Switch Statement	Tower of Hanoi Algorithm Game of Life Algorithm Inference Engine (A.I.)	switch.js ... pop_quiz.js
7	Control Structures II <u>Iteration Statements (Loop)</u> For Loop While Loop Do-While Loop For-in Loop Nested Loops Counter-controlled Loops	Data Authentication Multiplication Table Arbitrary Matrices Linear Search Selection Sort Fibonacci Algorithm	for.js while.js do_while.js for_in.js ... series.js nested_loop.js matrix.js tree.js
8	<u>Subprograms</u> Functions and Procedures Recursion Subroutines Parameters and Arguments Lambda Function Nested Subprograms Overloaded Subprograms Generics	Macroinstructions (Macros) Factorial Mean, Median, Mode Expert System (A.I.)	function.js recursion.js lambda.js subroutines.js
9	Abstract Data Types <u>Attributes of Object Oriented Programming</u> Class Members Properties Constructor Methods (Setter, Getter, Logic, Display) Instance of a Class Object of a Class <u>Concepts of Object Oriented Programming</u> Abstraction Encapsulation Inheritance Polymorphism Interface	Data Abstraction Static Properties and Methods Entity Relationship Model Class and Object Diagrams (UML) Automated Teller Machine (ATM)	encapsulation.js inheritance.js
10	<u>Built-in APIs</u> String Number Array	Application Programming Interface (API) String Buffer and String Builder Anagrams and Palindromes Virtual Assistant (A.I.)	table.js arguments.js timestamp.js ...

	Math Date Console <u>User-defined APIs (macros/*)</u> Buffer Binary Calendar MariaDB		salute.js palindrome.js anagram.js
11	Regular Expression ? Applications of Pattern Matching ? Data Sterilization v Data Authentication	Data Sterilization Pattern Recognition (Brute-Force) Character Encoding Place-value Notation	regex.js pattern.js encoding.js ... binary.js hexadecimal.js
12	Errors and Debugging <u>Types of Errors</u> Compile-time Errors Run-time Errors <u>Source of Errors</u> Syntax Errors Semantic Errors Logical Errors ? Common Sources of Errors (Vitamin A)	Software Testing <u>HTTP Status Codes</u> ✓ 100 (Continue), 102 (Processing) ✓ 200 (OK), 202 (Accepted) 204 (No Content) ✓ 300 (Multiple Choices), 302 (Found), 305 (Use Proxy) ✓ 400 (Bad Request), 401 (Unauthorized), 402 (Payment Required), 403 (Forbidden), 404 (Not Found) ✓ 500 (Server Error), 502 (Bad Gateway), 503 (Service Unavailable), 507 (Insufficient Storage)	exit.js debugger.js
13	<u>Exception Handling</u> NumberFormatException ArithmeticException ArrayIndexOutOfBoundsException ClassNotFoundException FileNotFoundException ? Difference between Underflow and Overflow	Software Quality Control	exception.js

MODULE 2: Source code editing with Visual Studio Code

#	LECTURE	SCOPE	EXERCISE
14	Workspace Environment Menu Bar Tabbed Bar Toolbox Navigation Pane Text-editor Frame Breadcrumbs* Scroll-spy Output/Terminal Window Status Bar	Integrated Development Environment (IDE) Text Editors Static Code Analysis Code Refactoring Debugging Tools	
15	<u>Shortcuts</u> (Ctrl + K Ctrl + S) <u>File</u> New (Ctrl + N) Open (Ctrl + O) Save (Ctrl + S) Save As (Ctrl + Shift + S) Folder (Ctrl + K Ctrl + O) Rename (F2) <u>Edit</u> Select All (Ctrl + A) Undo (Ctrl + Z) Redo (Ctrl + Y) Cut (Ctrl + X) Copy (Ctrl + C) Paste (Ctrl + V) Find and Replace (Ctrl + H) Toggle Line Comment (Ctrl + /) Toggle Block Comment (Shift + Alt + A) Highlight (Double Click) Quote (Highlight and Shift + 2) <u>View</u> Toggle Navigation Pane (Ctrl + B) Explorer (Ctrl + Shift + E) Search (Ctrl + Shift + F) Split (Ctrl + \) View Directory (Shift + Alt + R) Close (Ctrl + F4) Close All (Ctrl + K Ctrl + W)	# Misc ctrl + , (goto settings) ctrl + k ctrl + s (all shortcuts) ctrl + k ctrl + t (change theme) ctrl + alt + n (run code with Code Runner Extension) # File ctrl + n (new) ctrl + shift + n (new window) ctrl + o (open) ctrl + k ctrl + o (open folder) ctrl + s (save) ctrl + shift + s (save as... or clone file) ctrl + w (close) ctrl + k ctrl + w (close all) ctrl + f4 (close tab) alt + f4 (close window) # Edit ctrl + a (select all) ctrl + c (copy) ctrl + x (cut) ctrl + v (paste) ctrl + z (undo) ctrl + y (redo) ctrl + f (find only) ctrl + h (find & replace) ctrl + / (toggle line comment) shift + alt + a (toggle block comment)	# View ctrl + page-up (next tab) ctrl + page-down (previous tab) ctrl + shift + t (reopen closed tab) ctrl + tab (toggle tab) ctrl + shift + tab (toggle tab inverted) ctrl + shift + e (explorer toolbar) ctrl + shift + f (find toolbar) ctrl + shift + h (replace toolbar) ctrl + shift + x (my extensions) ctrl + \ (split editor) ctrl + + (zoom in) ctrl + - (zoom out) ctrl + ` (toggle terminal) ctrl + j (toggle terminal) ctrl + b (toggle toolbar) ctrl + e (goto file)

	<u>Go</u> Back (Alt + LEFT) Forward (Alt + RIGHT) Next (Ctrl + PgUp) Previous (Ctrl + PgDown) Switch (Ctrl + TAB, Ctrl + Shift + TAB) History (Ctrl + P) Source Line (Ctrl + G) <u>Manage</u> Settings (Ctrl + ,) Themes (Ctrl + K Ctrl + T)	double-click (highlight word) triple-click (highlight paragraph) alt + left (cursor trace backward) alt + right (cursor trace forward) alt + up (shift paragraph up) alt + down (shift paragraph down) shift + alt + up (duplicate paragraph up) shift + alt + down (duplicate paragraph down) alt + z (toggle word wrap)	ctrl + p (goto file) ctrl + g (goto line) f2 (rename directory) shift + alt + r (open file explorer)
16	<u>Terminal</u> (Ctrl + Shift + `) Root (cd\ View (dir) Navigate (TAB, Shift + TAB) Change (cd path\ Previous (cd.. Execute (ENTER) History (F7) Escape (ESC) Help (help) Clear (cls) Commands (cmd/?) Hidden (attrib -s -h -r /s /d) Close (Alt + F4)	<u>Windows Command Line (cmd)</u> execute command [enter] clear command [esc] command history [f7] previous command [arrow-up] next command [arrow-down] close prompt [alt + f4] help help clear cls commands cmd/? root directory cd\ parent directory cd.. open directory cd folder-name open sub-directory cd folder-name\sub-folder-name fwd-scan directory cd [tab] bwd-scan directory cd [shift + tab] rgx-scan directory cd folder-name-sub-string view directory dir new directory mkdir folder-name del directory rmdir folder-name show hidden files attrib -s -h -r /s /d * directory names are case-insensitive * can only delete empty directory	

MODULE 3: Advanced Data Structures with JavaScript Object Notation (JSON)			
#	LECTURE	SCOPE	EXERCISE
17	Introduction to JSON	Database Schema	students.json references.json

MODULE 4: Parallel Processing with Asynchronous JavaScript and XML (AJAX)

#	LECTURE	SCOPE	EXERCISE
18	Introduction to AJAX	Applications of Parallel Processing Threading Hypertext Transfer Protocol (HTTP) Web Services Simple Object Access Protocol (SOAP) RDF Site Summary (RSS) Search Engine Optimization (SEO) Instant Messaging (IM)	*students.js *references.js
19	Goto Statements	Unconditional Branching Basic Blocks	
20	Pointers	Compiler Design Language Translators Intermediate Code Generation (Indirect Triples)	
21	Process Scheduling Scheduling Algorithms First Come, First Serve (FCFS) Shortest Job First (SJF) Round-robin	CPU Scheduling Criteria Time Complexity	
22	Lexical Analysis ? Requirements traditionally imposed on Code Generators	Target Machine Code Lexical and Syntax Analysis Intermediate Code Generation (Quadruple)	
23	Instruction Cost ? Cost of a dot-product program	Memory Management Register Allocation	
24	Code Optimization ? Principal sources of Code Optimization	Peephole Optimization Memory Reallocation Overloaded Variable Declaration* Overloaded Variable Initialization* Parameterized Nesting* Lambda Return*	
25	Prefix, Infix and Postfix Notation	List Programming (LISP)	

MODULE 5: Server-side Scripting with Node.js Run-time Environment			
#	LECTURE	SCOPE	EXERCISE
26	Introduction to Server-side Scripting	Hypertext Transfer Protocol (HTTP) Common Gateway Interface (CGI)	scores_data.json scores_main.js
27	File Upload	File Transfer Protocol (FTP) Data Encryption	upload.js
28	File System	Folder Management System	stream.js
29	Email System	Computer Communication System Multipurpose Internet Mail Extensions (MIME)	email.js
30	Sessions	Computer Network Security	sessions.js

MODULE 6: Document-oriented Database Management with MongoDB

#	LECTURE	SCOPE	EXERCISE
31	<u>Database Management System</u> Database Table Fields Rows Records Relationship Cardinality <u>Field Data Types</u> INT CHAR VARCHAR ENUM SET TIMESTAMP (yyyy-mm-dd hh:mm:ss) BLOB ? Difference between DBMS and RDBMS ? Unique Key, Primary Key and Foreign Key	Database Management System (DBMS) Entity Relationship Diagram (ERD) Persistent/Non-volatile Storage Binary Large Object (BLOB)	schema.doc
32	<u>Basic SQL Operations</u> Create (INSERT INTO, CREATE) Read (SELECT FROM) Update (UPDATE...SET, ALTER) Delete (DELETE FROM, DROP)	Structured Query Language (SQL) Data Query Language (DQL) Data Definition Language (DDL) Data Manipulation Language (DML) Representational State Transfer (REST)	database.js
33	<u>Advanced SQL Commands</u> Search (WHERE...AND/OR) Sort (ORDER BY: ASC, DESC) Filter (DISTINCT, LIMIT)	Searching Algorithms Sorting Algorithms	

MODULE 7: Software Development Life Cycle (SDLC) using Waterfall Model and MVC			
#	LECTURE	SCOPE	EXERCISE
34	Requirement Phase	Software Development Life Cycle (SDLC) Systems Analysis & Design Facts Finding Techniques Cost Estimation Models	waterfall.doc cocomo.js
35	Design Phase	Business Process Modelling (BPM) Unified Modelling Language (UML) Entity Relationship Diagrams (ERD)	usecase.doc
36	Implementation Phase (Development)	Class and Object Diagrams	
37	Testing Phase	Software Testing Life Cycle (STLC)	
38	Deployment Phase	Secure Hypertext Transfer Protocol (HTTPS) Uniform Resource Locator (URL) Website Domain and Hosting Domain Name and Extension Internet Service Providers (ISPs) Web Servers and Cloud Storage ... Change Over Methods	
39	Maintenance Phase ? Why do systems change	Software Configuration Management Version Control Systems (VCS) Distributed Revision Control Systems (DRCS) Git Repository	
40	Software Documentation	Requirements Documentation Architecture Design Documentation Technical Documentation End-User Documentation Marketing Documentation	documentation.doc

MODULE 8: Software Prototyping with Hypertext Markup Language (HTML5)			
#	LECTURE	SCOPE	EXERCISE
41	Introduction to Web Design Web Design Glossary Classification of Websites Careers in Web Design & Development ? Enterprise and Distributed Systems	Website Architecture Content Management System (CMS)	
42	HTML5 Program Structure Tags Attributes Elements Meta Site Folder	Semantic Web	clipboard.html
43	Turbo Tags Basic Formatting Tags Hyperlinks Style Attribute Special Characters	Cognitive Science Case-based Reasoning (CBR) ... Literature Review (References) American Psychological Association (APA) American Medical Association (AMA) Council of Science Editors (CSE) ... CSS Display Property	
44	Window Location Navigator Threading	Browser Object Model (BOM)	resize.html print.html ... countdown.html loading.html
45	Events Event Listeners Event Handlers ? The impact of <i>Gamification</i> in E-Learning	Document Object Model (DOM) Rich Text Editors ... Game Development Interactive Prototyping Bouncing Ball Algorithm	onclick.html onchange.html onkeyup.html ... jumble_word.html
46	Heading Paragraph, Line Break, Horizontal Rule Image Footer Address Article	<u>MVC Architecture</u> model (class_*.js, data_*.json) view (style_*.css) ctrl (server_*.js)	<u>ZentrySME/*</u> index.html about.html

47	Forms Form Validation API HTTP POST Sessions Time		login.html register.html stocks_add.html sales_add.html settings.html
48	Header Nav Main HTTP GET Lists	Enterprise Resource Planning (ERP)	dashboard.html sales_preview.html
49	Tables	Inventory Management System Business Intelligence	stocks_record.html sales_record.html sales_report.html accounts.html
APX	MONETIZATION ? Business Model vs. Revenue Model ? User-base vs. Consumer-base ? Organic vs. Inorganic Growth ? Minimum Viable Product (MVP) ? The Real Product-Market Fit_ (MPF) <u>BLEND Design</u> User Interface (UI) User Experience (UX) User Engagement (UE) <u>SWORD Web Design</u> Semantic Responsive Progressive <u>Revenue Markup Language (RXML)</u> 1. Advert Placement 2. Classifieds 3. Direct Sales 4. Drop Shipping 5. Enterprise Services 6. <i>Freemium</i> Model 7. Multi-level Marketing 8. Rental/Leasing 9. Sponsored Content 10. Subscription Service	<u>Website Classification</u> 1. Academic 2. Betting 3. Blog 4. Classifieds 5. Cloud Service 6. Corporate or Business 7. Directory or Repository (all file formats) 8. E-Commerce or Marketplace 9. Enterprise (SaaS/PaaS/IaaS) 10. Forum/Message Boards 11. Gaming 12. News/Magazine 13. Non-profit 14. Portal 15. Portfolio 16. Search Engine 17. Storage Service 18. Streaming Service 19. Social Media or Social Network 20. Wiki- <u>User Metrics (Visitors < Users < Impressions)</u> Monthly Active Users (MAU) Session Rate Conversion Rate Retention Rate	<u>Marketing Mix (7P's)</u> Product Price Place Promotion People Process Physical Space <u>SWOT Analysis</u> Strengths Weaknesses Opportunities Threats <u>* As A Service</u> Software (SaaS) Platform (PaaS) Infras. (IaaS) Backend (BaaS)

EXAMINATION & CERTIFICATION

FOREWORD

Two core **attributes** of a Maven programmer are Confidence and Perseverance. A Maven programmer exhibits certain **behaviours** when writing code; Algorithm Design, Semantic Nomenclature, Identifier Hoisting, Code Readability (tidy) and Code Optimization. Thus, the 3 Ps of programming are Practice! Practice!! Practice!!!

...

“use strict”;

class Maven

{

 constructor (agr1, agr2)

 {

this.confidence = agr1;

this.perseverance = agr2;

 }

get algorithmDesign() {**return** true;}

get semanticNomenclature() {**return** true;}

get identifierHoisting() {**return** true;}

set codeReadability (tidy) {**window**.tidy = true;}

get codeOptimization() {**return** true;}

thus = () => {**for** (**let** i = 1; i <= 3; i++) **alert**(“Practice!!!”);}

}

// TODO Express the illustration above using object-oriented programming

const programmer = **new** Maven(true,true);

programmer.thus();

REFERENCES

- Brendan Eich, creator of JavaScript scripting language 1995, Netscape Communications Corporation, Mozilla Foundation, ECMA International, <https://www.javascript.com/>.
- Douglas Crockford, author of JavaScript Object Notation (JSON) 2000, Paypal, <https://www.json.org/>.
- Emmanuel Tugbeh, author of SlingshotJS framework for Masters in Computer Science 2019, SlingshotJS Working Group, HWP Labs, <http://www.hwplabs.com/>.
- Google LLC., owner of Google Chrome 2008, <https://www.google.com/chrome/>. Creator of Asynchronous JavaScript and XML (AJAX) 2004, coined by in Jesse James Garrett in 2005, standardized by the World Wide Web Consortium (W3C) in 2004, <https://www.w3c.org/>.
- Herbert Benington, author of Waterfall SDLC Model 1956, Symposium on Advanced Programming Methods for Digital Computers, <https://www.worldcat.org/oclc/10794738/>.
- Isaac Schlueter, developer of Node Package Manager (NPM) 2010, Joyent Inc., <https://www.npmjs.com/>.
- Microsoft Corporation, owner of Visual Studio Code Integrated Development Environment (IDE) 2015, <https://code.visualstudio.com/>.
- Ryan Dahl, creator of the Node.js JavaScript run-time environment 2009, Joyent Inc., <https://www.nodejs.org/>.
- Tim Berners-Lee, developer of Hypertext Markup Language (HTML) 1993, Web Hypertext Application Technology Working Group (WHATWG), <https://www.whatwg.org/>.
- Trygve Reenskaug, author of Model-View-Controller (MVC) 1970, Xerox Palo Alto Research Center (PARC), Microsoft Corporation, <https://dotnet.microsoft.com/apps/mvc/>.